

Data Dictionary

Project: MoFlow: Smart Recurring Bill Payments Platform

Document Type: Data Dictionary

Role: Business Analyst

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1. Introduction

1.1 Purpose

This Data Dictionary defines the key business data elements used within the MoFlow Smart Recurring Bill Payments Platform. It provides a common understanding of the data stored, processed, and exchanged across the application.

The document supports business analysis, system design, database development, API integration, testing, and reporting by ensuring consistency in data definitions.

1.2 Scope

The Data Dictionary covers the primary business entities within MoFlow, including:

- User
- Mobile Money Account
- Biller
- Recurring Payment
- Payment Transaction
- Notification
- Administrator
- Audit Log

2. User Entity

Field Name	Data Type	Length	Required	Description	Example
User ID	UUID	36	Yes	Unique identifier for each user	U-10001
First Name	Text	50	Yes	User's first name	Reginald
Last Name	Text	50	Yes	User's surname	Erzoah
Email Address	Email	100	Yes	Registered email address	reginald@email.com
Mobile Number	String	15	Yes	Registered mobile number	+233547334867
Password Hash	String	255	Yes	Encrypted password	*****
Date of Birth	Date	-	No	User's date of birth	2000-05-10
Account Status	Enum	-	Yes	Active, Suspended, Inactive	Active
Created Date	DateTime	-	Yes	Account creation date	2026-07-20
Last Login	DateTime	-	No	Last successful login	2026-07-25

3. Mobile Money Account Entity

Field Name	Data Type	Length	Required	Description	Example
Account ID	UUID	36	Yes	Unique account identifier	ACC-001
User ID	UUID	36	Yes	Linked user	U-10001
Provider	Enum	-	Yes	Mobile money provider	MTN MoMo
Account Number	String	15	Yes	Mobile money number	0547334867
Account Name	Text	100	Yes	Registered account name	Reginald Erzoah
Verification Status	Enum	-	Yes	Verified, Pending, Failed	Verified
Date Linked	DateTime	-	Yes	Date account was linked	2026-07-20

4. Biller Entity

Field Name	Data Type	Length	Required	Description	Example
Biller ID	UUID	36	Yes	Unique biller identifier	BILL-001
Biller Name	Text	100	Yes	Service provider name	ECG Ghana
Category	Enum	-	Yes	Utility, Rent, Internet, School Fees, Insurance	Utility
Account Reference	String	50	Yes	Customer account with biller	ECG998877
Status	Enum	-	Yes	Active, Inactive	Active

5. Recurring Payment Entity

Field Name	Data Type	Length	Required	Description	Example
Schedule ID	UUID	36	Yes	Unique recurring payment identifier	SCH-001
User ID	UUID	36	Yes	Owner of schedule	U-10001
Biller ID	UUID	36	Yes	Linked biller	BILL-001
Payment Amount	Decimal	10,2	Yes	Scheduled payment amount	250.00
Currency	String	3	Yes	Payment currency	GHS
Frequency	Enum	-	Yes	Daily, Weekly, Monthly, Quarterly, Annually	Monthly
Start Date	Date	-	Yes	First payment date	2026-08-01
End Date	Date	-	No	Final payment date	2027-08-01
Next Payment Date	Date	-	Yes	Upcoming payment date	2026-09-01

Schedule Status	Enum	-	Yes	Active, Paused, Cancelled	Active
Reminder Enabled	Boolean	-	Yes	Reminder preference	Yes
Created Date	DateTime	-	Yes	Schedule creation date	2026-07-20

6. Payment Transaction Entity

Field Name	Data Type	Length	Required	Description	Example
Transaction ID	UUID	36	Yes	Unique payment transaction	TXN-001
Schedule ID	UUID	36	Yes	Linked recurring payment	SCH-001
Payment Amount	Decimal	10,2	Yes	Paid amount	250.00
Transaction Date	DateTime	-	Yes	Payment execution time	2026-08-01 09:00
Payment Status	Enum	-	Yes	Pending, Processing, Successful, Failed	Successful
Reference Number	String	50	Yes	Provider reference	MOMO56789
Failure Reason	Text	255	No	Reason for failure	Insufficient Funds

7. Notification Entity

Field Name	Data Type	Length	Required	Description	Example
Notification ID	UUID	36	Yes	Unique notification identifier	NOT-001
User ID	UUID	36	Yes	Notification recipient	U-10001

Notification Type	Enum	-	Yes	Reminder, Success, Failure	Reminder
Message	Text	255	Yes	Notification content	Payment due tomorrow
Delivery Channel	Enum	-	Yes	SMS, Email, Push	Push
Delivery Status	Enum	-	Yes	Sent, Delivered, Failed	Delivered
Sent Date	DateTime	-	Yes	Date notification sent	2026-07-31

8. Administrator Entity

Field Name	Data Type	Length	Required	Description	Example
Admin ID	UUID	36	Yes	Administrator identifier	ADM-001
Full Name	Text	100	Yes	Administrator name	Jane Doe
Email Address	Email	100	Yes	Official email	admin@moflow.com
Role	Enum	-	Yes	Administrator, Support, Operations	Operations
Status	Enum	-	Yes	Active, Inactive	Active

9. Audit Log Entity

Field Name	Data Type	Length	Required	Description	Example
Log ID	UUID	36	Yes	Audit record identifier	LOG-001
User ID	UUID	36	Yes	User performing action	U-10001
Activity	Text	255	Yes	Recorded activity	Created Recurring Payment

Timestamp	DateTime	-	Yes	Date and time of activity	2026-07-20 14:15
IP Address	String	45	No	User IP address	102.xxx.xxx.xxx
Device Type	Text	50	No	Device used	Android

10. Enumerated Values

Account Status

Value	Description
Active	User can access all features
Suspended	Access temporarily disabled
Inactive	Account no longer active

Schedule Status

Value	Description
Active	Payments will execute automatically
Paused	Payments temporarily suspended
Cancelled	Schedule permanently stopped

Payment Status

Value	Description
Pending	Waiting to be processed
Processing	Currently being processed
Successful	Payment completed successfully
Failed	Payment unsuccessful
Cancelled	Payment cancelled before processing

Notification Types

Value	Description
Reminder	Upcoming payment reminder
Success	Successful payment confirmation
Failure	Failed payment notification
System	General platform updates

Payment Frequency

Value	Description
Daily	Every day
Weekly	Every week
Monthly	Every month
Quarterly	Every three months
Annually	Once every year

11. Data Relationships

Parent Entity	Child Entity	Relationship
User	Mobile Money Account	One-to-Many
User	Recurring Payment	One-to-Many
User	Notification	One-to-Many
User	Audit Log	One-to-Many
Biller	Recurring Payment	One-to-Many
Recurring Payment	Payment Transaction	One-to-Many
Administrator	Audit Log	One-to-Many

12. Data Governance Guidelines

- Every entity shall have a unique identifier (UUID).
- Personally identifiable information (PII) shall be encrypted where applicable.
- Mandatory fields must be validated before data is saved.
- Audit logs shall be retained according to organizational policies.
- Access to sensitive information shall be restricted based on user roles.
- Payment data shall comply with applicable financial and data protection regulations.

13. Conclusion

The MoFlow Data Dictionary establishes a standardized definition of the platform's core business data. By documenting entities, attributes, data types, validation requirements, and relationships, it promotes consistency across business analysis, solution design, development, API integration, testing, reporting, and ongoing maintenance. It serves as a single source of truth for all stakeholders working on the MoFlow platform.